

SHDragron

Undergraduate Researcher in Autonomous Driving Perception

📍 South Korea 🔗 shdragon

Summary

Undergraduate researcher and vehicle engineer working on autonomous driving perception, robust BEV representation, multi-camera calibration, and real-vehicle deployment. My background connects Formula Student vehicle engineering with computer vision research for autonomous driving systems.

Experience

Formula Student, Mechanical Analysis Lead Engineer Led and contributed to vehicle design, manufacturing, structural analysis, and optimization for Formula Student vehicles. • Led design, manufacturing, and optimization work for the Forsythia E-Formula Vehicle, Korea's first vehicle featuring a 4WD in-wheel motor drive system with a full monocoque chassis. • Contributed to the design and assembly of the RH27 monocoque-based vehicle. • Reduced component weight through optimization, including upright 1.3 kg to 0.9 kg and A-arm 9.8 kg to 7.5 kg. • Optimized mechanical systems using CATIA and Altair HyperWorks. • Conducted structural, thermal, and stress analyses for key vehicle components. • Collaborated on carbon composite monocoque and nose cone production using fiber-glass molds and autoclave processing.	South Korea 2020 – 2024 4 years
Computer Vision for Autonomous Driving, Undergraduate Researcher - Robust BEV Perception Researched multi-camera calibration and viewpoint-robust BEV semantic segmentation for autonomous driving. • Conducted research on multi-camera calibration and multi-view BEV semantic segmentation. • Designed and evaluated viewpoint-robust BEV perception pipelines using real-world driving data and CARLA simulation. • Analyzed the impact of camera viewpoint changes and calibration errors on BEV model performance. • Implemented and trained Cross View Transformer (CVT)-based models for multi-camera perception. • Awarded Grand Prize for graduation thesis and poster presentation.	South Korea 2025 – 2025 1 year
Computer Vision for Autonomous Driving, Undergraduate Researcher - Driver Behavior Analysis Studied driver behavior analysis using real-vehicle sensor data and weakly supervised time-series learning. • Built an integrated real-vehicle data acquisition environment using CAN, RT2000, PC, and eCon Systems cameras for driver behavior analysis. • Applied weakly supervised time-series classification baselines and proposed a novel architecture for multiple-instance time-series driver behavior analysis. • Authored a research paper on driver behavior analysis.	South Korea 2025 – 2025 1 year

T-CAR Project , Camera Perception Engineer Contributed to camera perception work for a real-vehicle autonomous driving platform. - Performed camera-LiDAR calibration and hardware synchronization. - Deployed StreamPETR as the 3D object detection model on the vehicle. - Applied TensorRT optimization for real-vehicle deployment.	South Korea 2025 – 2025 1 year
Formula Student Driverless , Team Lead Leading driverless Formula Student work from simulator-based perception to real-vehicle preparation. - Led LiDAR-based perception for the 1st Korean Formula Student Driverless Competition based on the Formula Student Driverless Simulator. - Won 1st place in the 1st Korean Formula Student Driverless Competition. - Preparing for the 2nd Formula Student Driverless Competition with a real-vehicle-based project.	South Korea 2025 – present 1 year
Computer Vision for Autonomous Driving , Undergraduate Researcher - Ongoing Research Preparing research projects on real-vehicle data acquisition, robust 3D object detection, and LiDAR-based reasoning. - Preparing a real-vehicle data acquisition project. - Studying LiDAR-camera based unsupervised 3D object detection for robustness. - Exploring LiDAR LLM approaches for ground reasoning.	South Korea 2026 – present 1 year

Awards

1st Place Winner — 1st Korean Formula Student Driverless Competition Team lead and LiDAR-based perception contributor in a simulator-based Formula Student Driverless competition.	2025
Grand Prize — Graduation Thesis and Poster Presentation Awarded for work on multi-camera calibration and multi-view BEV semantic segmentation for autonomous driving.	2025
Bronze Prize — Formula Student Korea Podium finish.	2024
Gold Prize — Altair Optimal Design Award Awarded for engineering optimization work.	
Silver Prize — KSAE Engineering Design Award Engineering design award from KSAE.	2020
35th Place — FSAE California International Formula Student competition result.	

Projects

Multi-camera Calibration and Multi-view BEV Semantic Segmentation Research project on robust BEV perception under camera viewpoint changes and calibration errors. - Used real-world driving data and CARLA simulation. - Implemented and trained CVT-based multi-camera perception models. - Evaluated viewpoint robustness and calibration-error sensitivity.

Multiple Instance Time Series for Driver Behavior Analysis

Driver behavior analysis project using integrated real-vehicle sensor data.

- Built a data acquisition system using CAN, RT2000, PC, and eCon Systems cameras.
- Applied weakly supervised time-series classification baselines.
- Proposed a novel architecture and authored a research paper.

T-CAR Real-Vehicle Camera Perception

Camera perception work for a real-vehicle autonomous driving platform.

- Camera-LiDAR calibration and hardware synchronization.
- StreamPETR deployment for 3D object detection.
- TensorRT optimization for onboard inference.

Forsythia E-Formula Vehicle

Formula Student vehicle featuring a 4WD in-wheel motor drive system and full monocoque chassis.

- Led mechanical analysis, design optimization, and manufacturing contributions.
- Contributed to carbon composite monocoque and nose cone production.

Formula Student Driverless

Driverless Formula Student work spanning LiDAR perception, simulator competition, and real-vehicle preparation.

- Led the team in the 1st Korean Formula Student Driverless Competition.
- Responsible for LiDAR-based perception.
- Won 1st place.

Ongoing Research Projects

Research directions in preparation for 2026 and beyond.

- Real-vehicle data acquisition project.
- LiDAR-camera based unsupervised 3D object detection for robustness.
- LiDAR LLM for ground reasoning.

Skills

Programming

Machine Learning and Computer Vision

Autonomous Driving Systems

Mechanical Design and Analysis

Interests

Robust Autonomous Driving Perception

Real-Vehicle Deployment